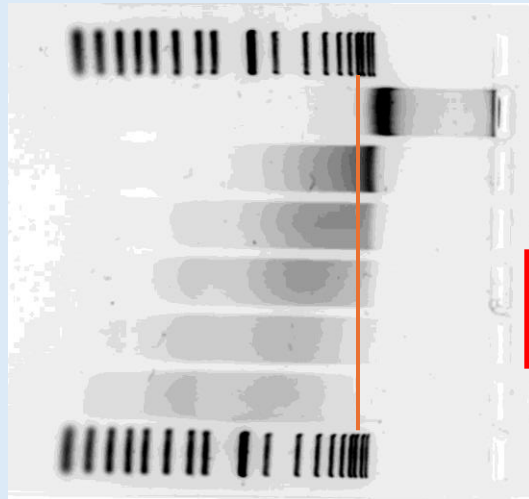


Fixed time (1 h), increasing FS mix (UltraExpress)



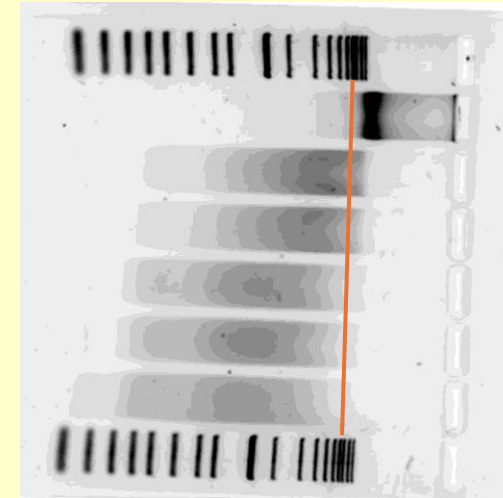
- 1 kb+
- 0
- 1
- 2 FS mix, uL
- 3 37°C 1250 rpm
- 4
- 5
- 1 kb+

7 kbp

A08 apple bead-bound chromatin

8-9/06/2026

Fixed FS mix (Ultra II, 2 uL), increasing incubation time



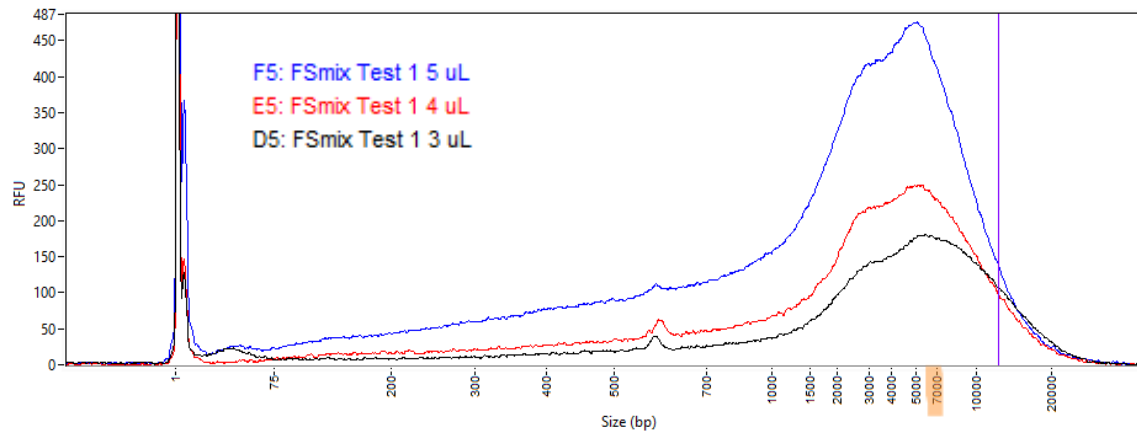
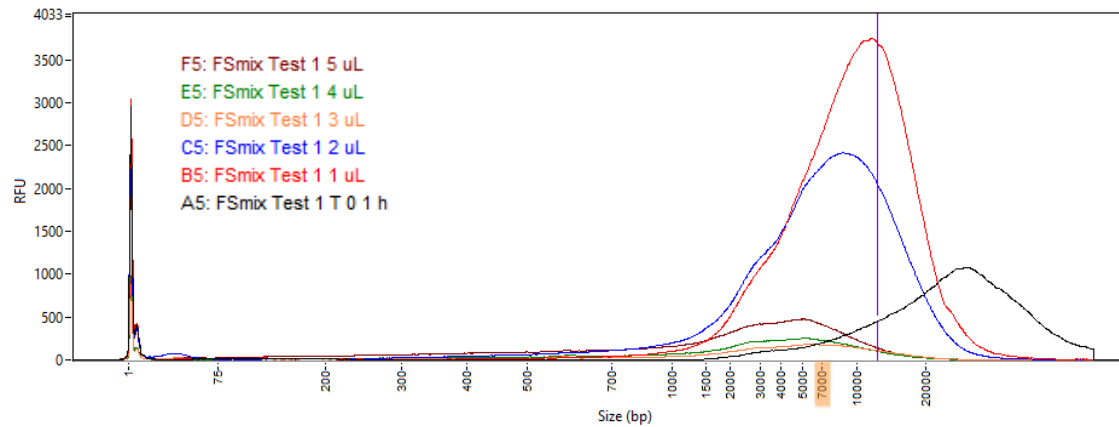
- 1 kb+
- 0 (3 h, no FS mix)
- 1 ≠ h, 37°C 1250 rpm
- 1.5
- 2
- 2.5
- 3
- 1 kb+

7 kbp

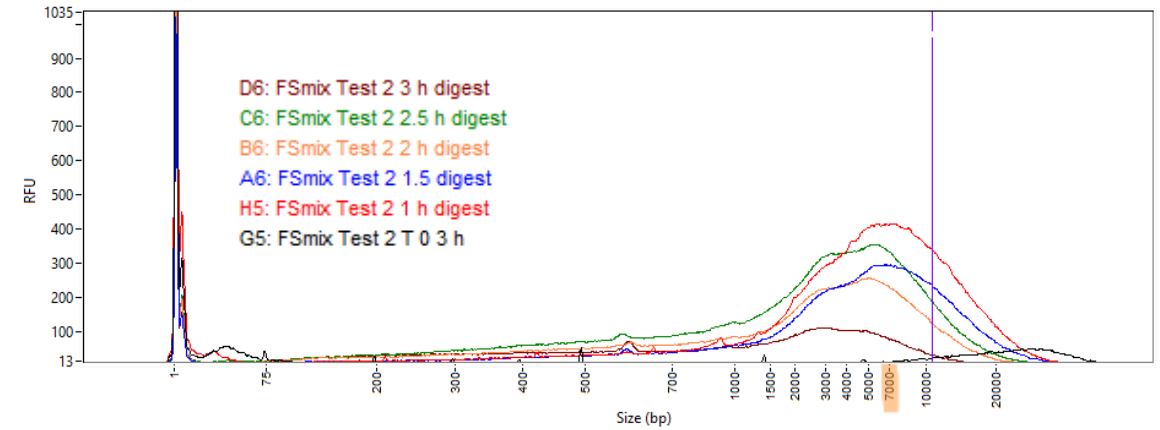
A02 apple bead-bound chromatin

16/06/2026

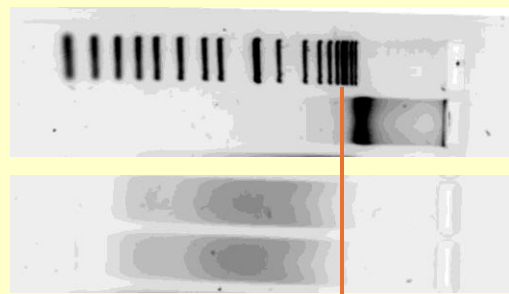
Fixed time (1 h), increasing FS mix



Fixed FS mix (2 uL), increasing incubation time



Fixed FS mix (2 uL), increasing incubation time



- 1 kb+
 - No FS mix, 3 h
 - 2 h
 - 2.5 h
- 37°C 1250 rpm

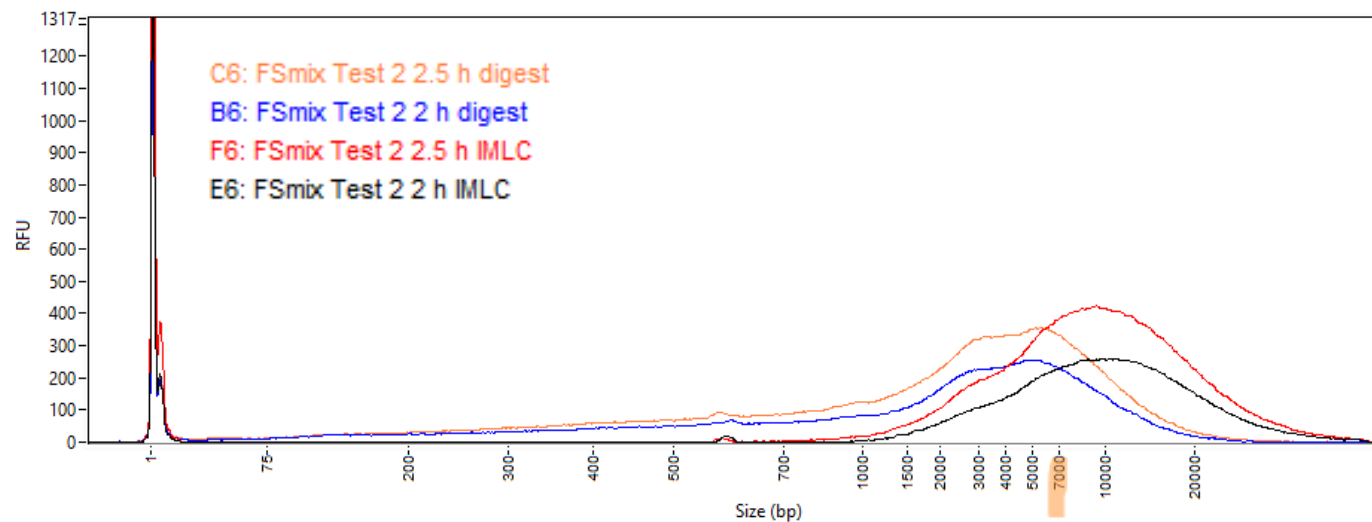
7 kbp

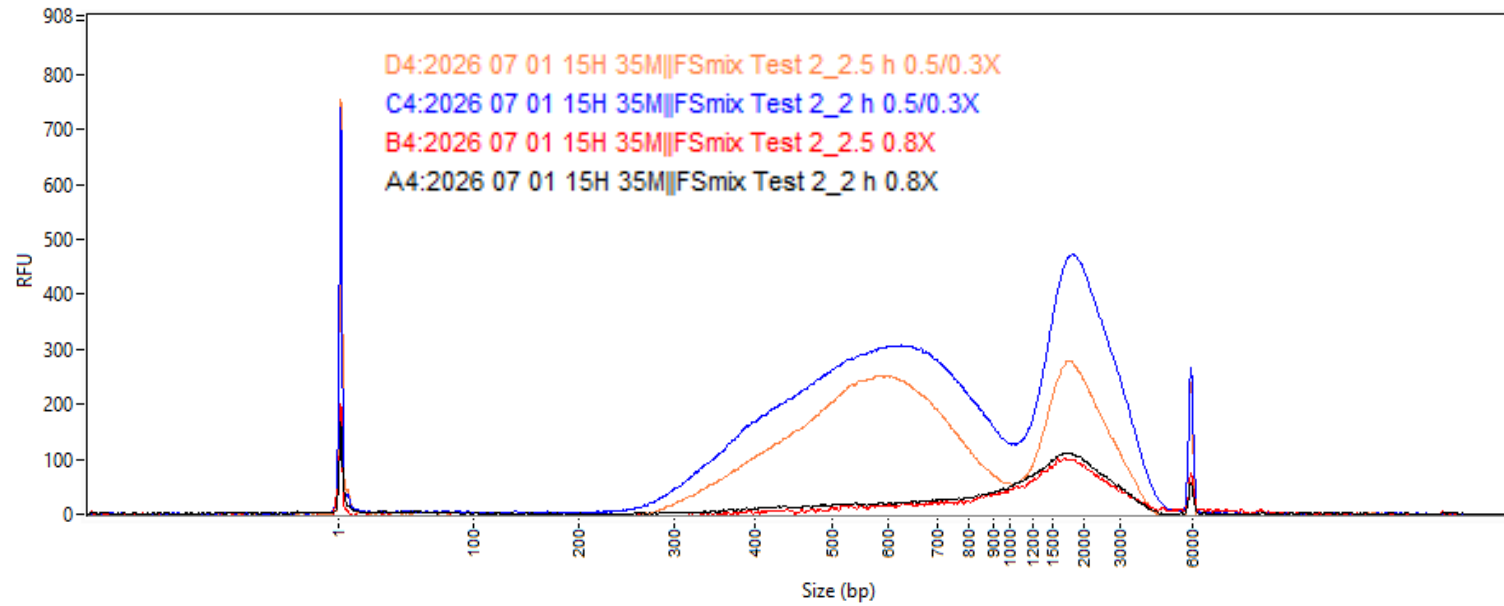
IMLC



- 2 h
- 2.5 h

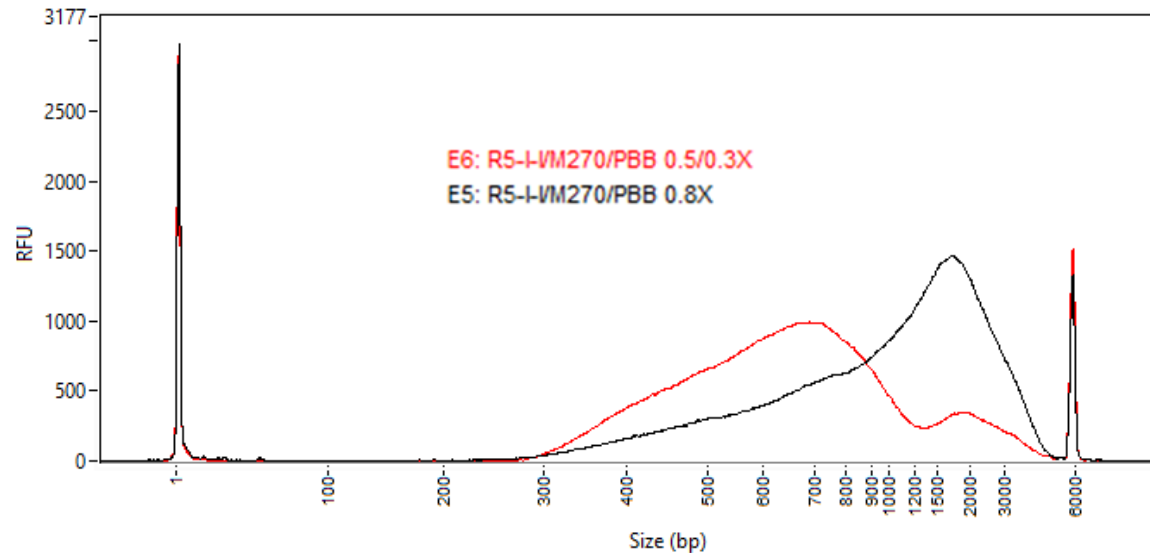
7 kbp





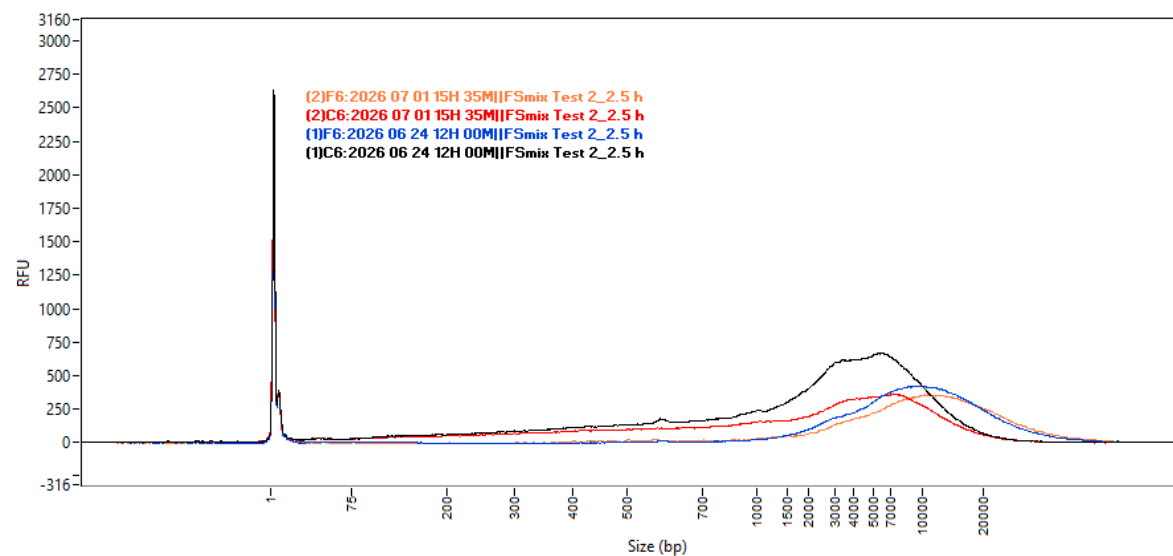
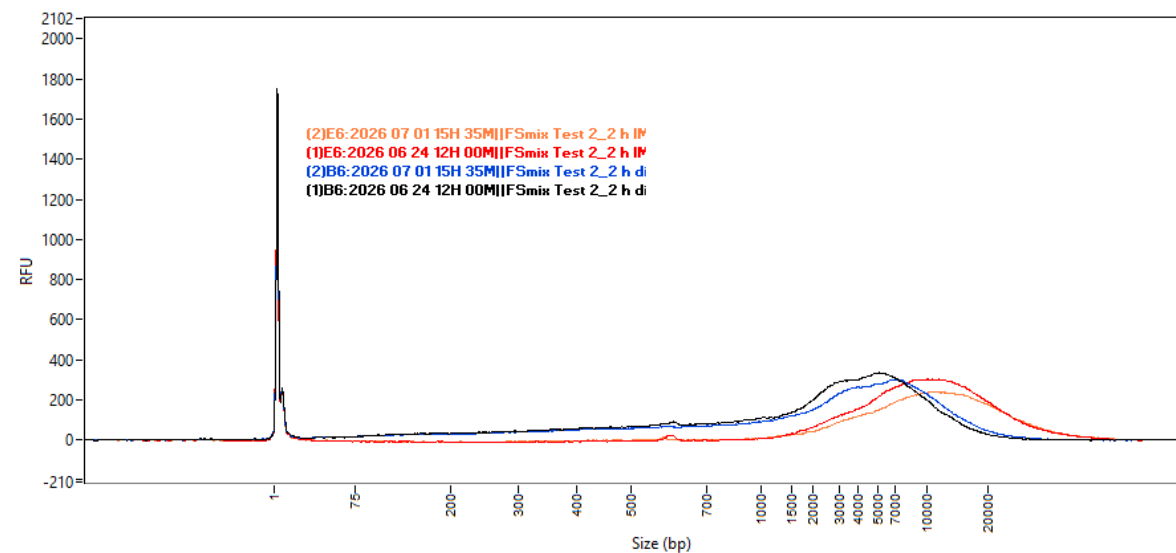
A bit too lumpy for the sequencer...

There is a big difference between profiles of the FS ns on the gel and on the FA. Seems like after random cutting, end repair and dA tailing, some aggregation occurred. And digestion there were many days between the gel and the FA run.



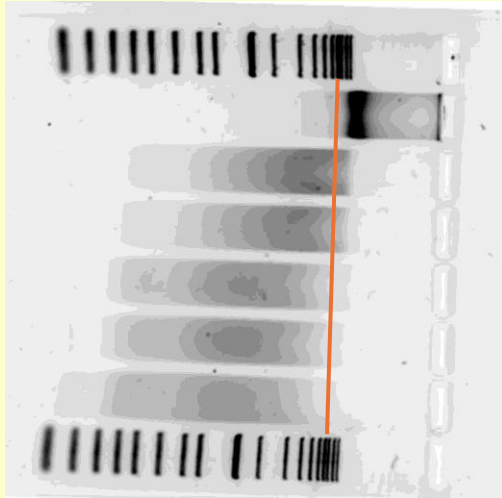
But the Intramolecular ligation happened right after the digestion (set up overnight on 17/06/2026)

The peak at ~ 2 kbp has been seen before but the ratio vs the lower peak is less than half its height



Fixed FS mix (2 uL), increasing incubation time

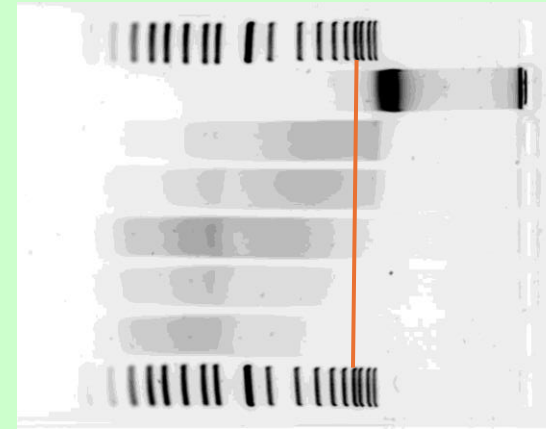
A02 apple bead-bound chromatin 16/06/2026



- 1 kb+
- 0 (3 h, no FS mix)
- 1 ≠ h, 37°C 1250 rpm
- 1.5
- 2
- 2.5
- 3
- 1 kb+

7 kbp

Same samples ran on 13/07/2026



- 1 kb+
- 0 (3 h, no FS mix)
- 1
- 1.5
- 2 ≠ h, 37°C 1250 rpm
- 2.5
- 3
- 1 kb+

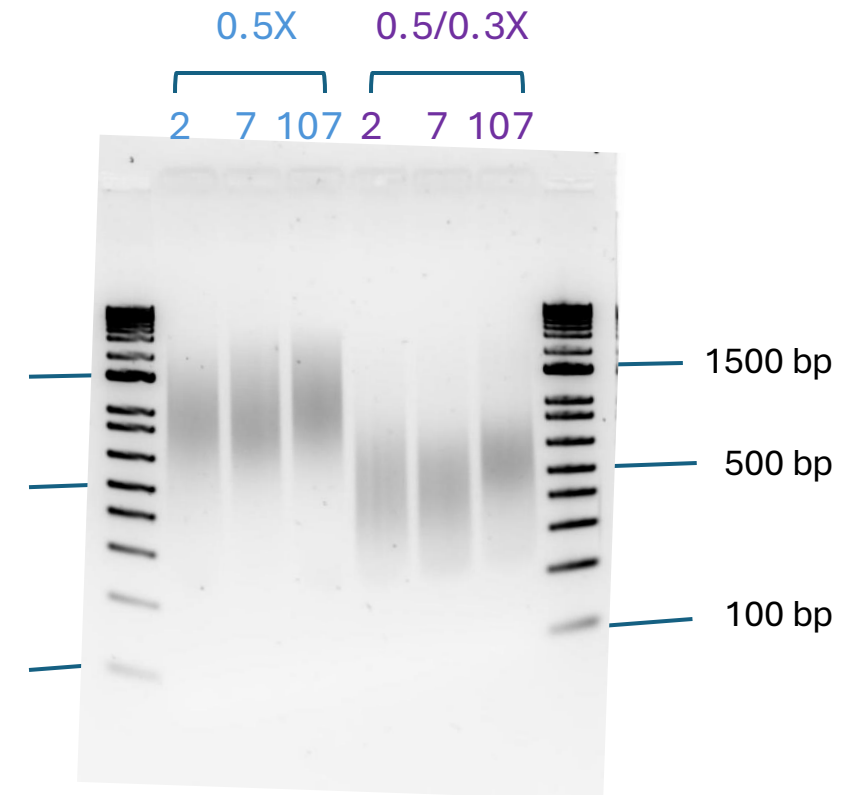
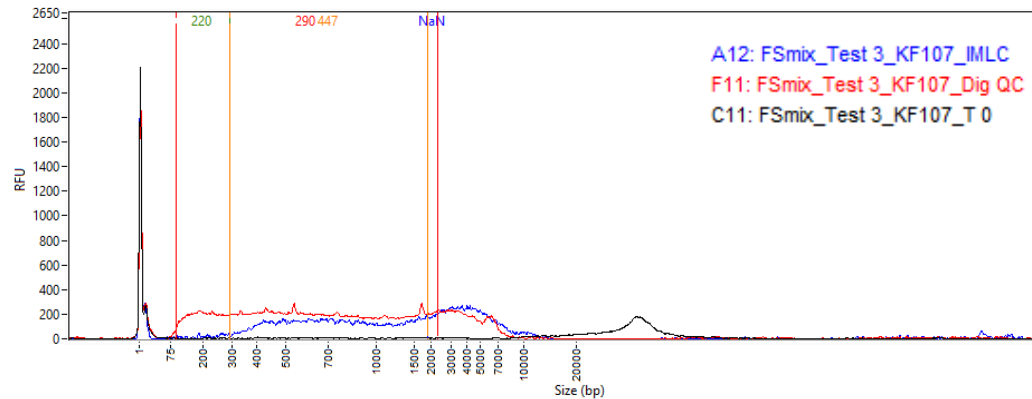
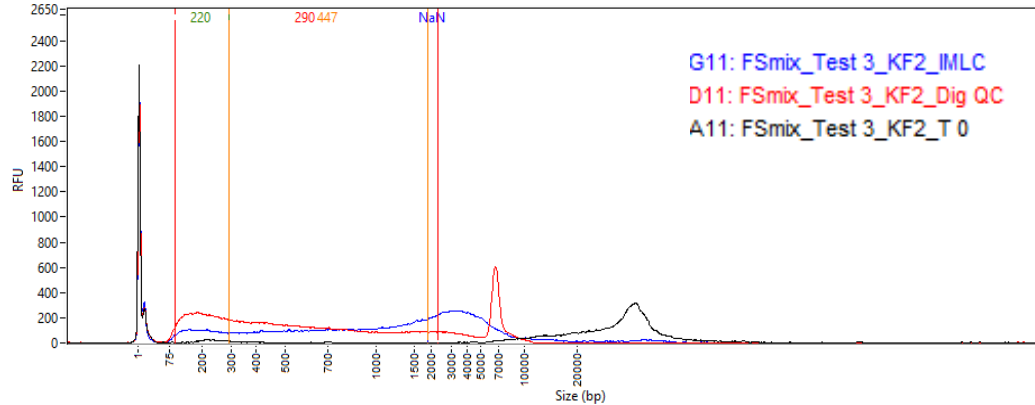
7 kbp

FSmix Test 3

Kiwifruit nuclei, 2 uL FS mix, 3 h @ 37°C 1250 rpm

Row Labels	Optimal 100 bp to 2500 bp	Low end 100 bp to 300 bp	High end 2000 bp to 2500 bp	Middle 300 bp to 2000 bp
FSmix_Test 3_KF107_Dig QC	81.7	20.2	3	58.8
FSmix_Test 3_KF107_IMLC	67.5	3.2	4.5	60
FSmix_Test 3_KF107_T 0	9.5	1.5	0.4	7.7
FSmix_Test 3_KF2_Dig QC	78.1	29	1.6	47.6
FSmix_Test 3_KF2_IMLC	67.4	14.4	4.7	48.7
FSmix_Test 3_KF2_T 0	5.3	3.6	0	1.6

Amplified libraries, 12 cycles,
double size selected, ~110 ng/lane

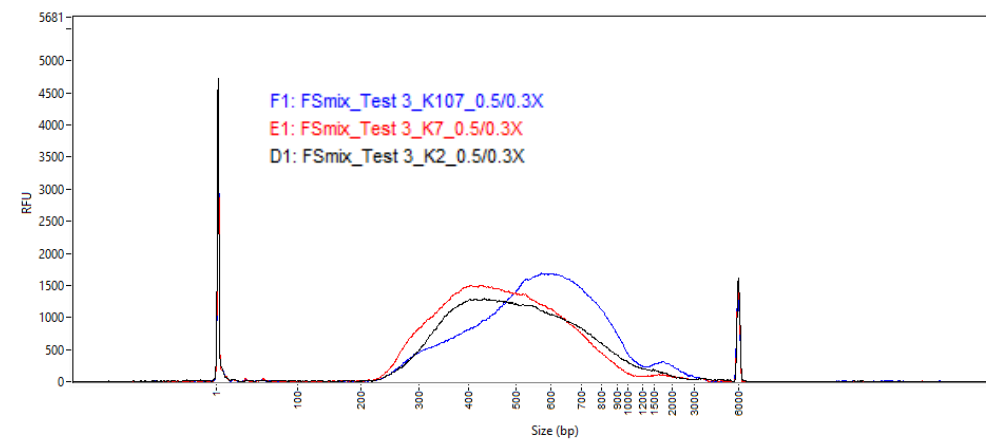
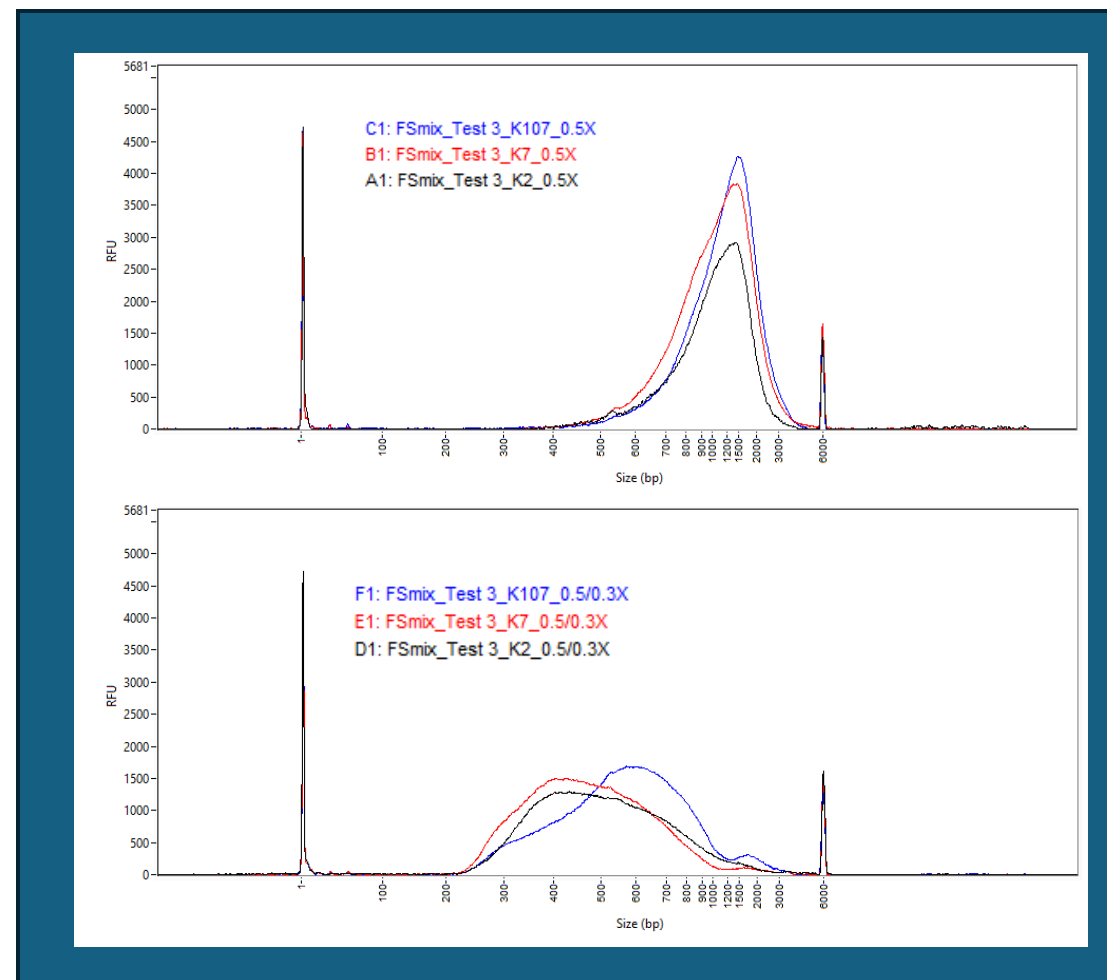
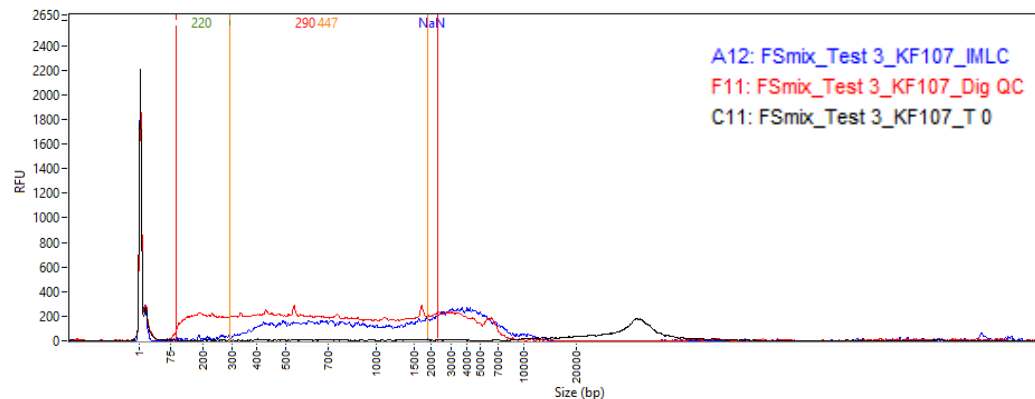
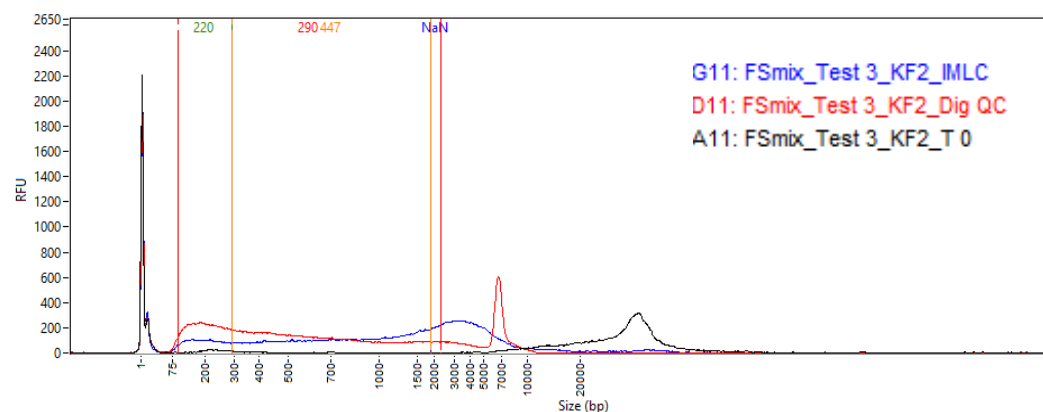


Note: the Fragment Analyzer traces of sample 7 were not analyzed because the digestion QC baseline can't be adjusted

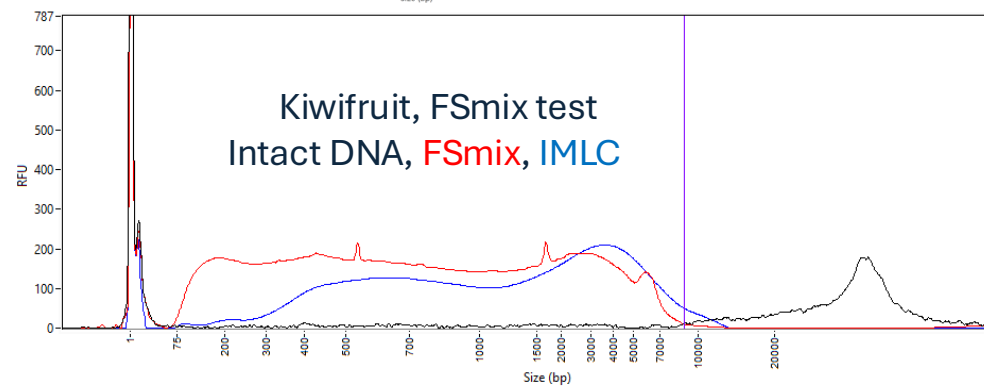
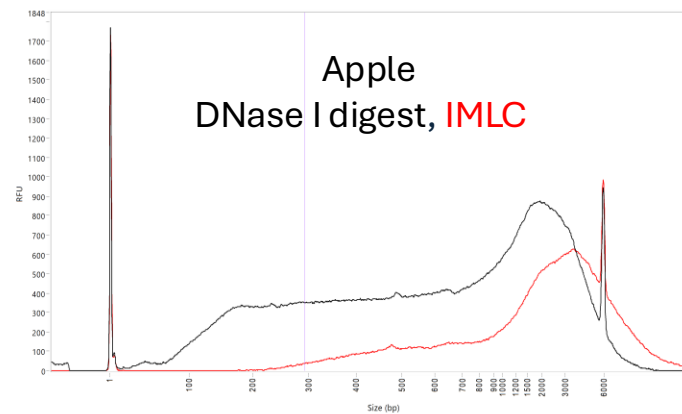
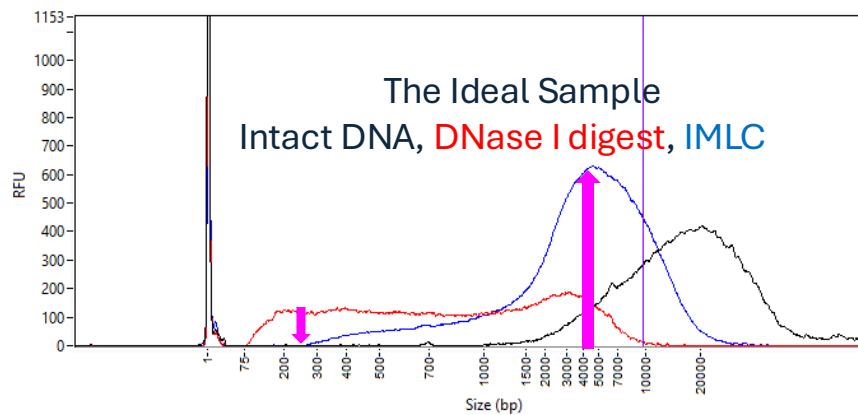
FSmix Test 3

Kiwifruit nuclei, 2 uL FS mix, 3 h @ 37°C 1250 rpm

Row Labels	Optimal 100 bp to 2500 bp	Low end 100 bp to 300 bp	High end 2000 bp to 2500 bp	Middle 300 bp to 2000 bp
FSmix_Test 3_KF107_Dig QC	81.7	20.2	3	58.8
FSmix_Test 3_KF107_IMLC	67.5	3.2	4.5	60
FSmix_Test 3_KF107_T 0	9.5	1.5	0.4	7.7
FSmix_Test 3_KF2_Dig QC	78.1	29	1.6	47.6
FSmix_Test 3_KF2_IMLC	67.4	14.4	4.7	48.7
FSmix_Test 3_KF2_T 0	5.3	3.6	0	1.6

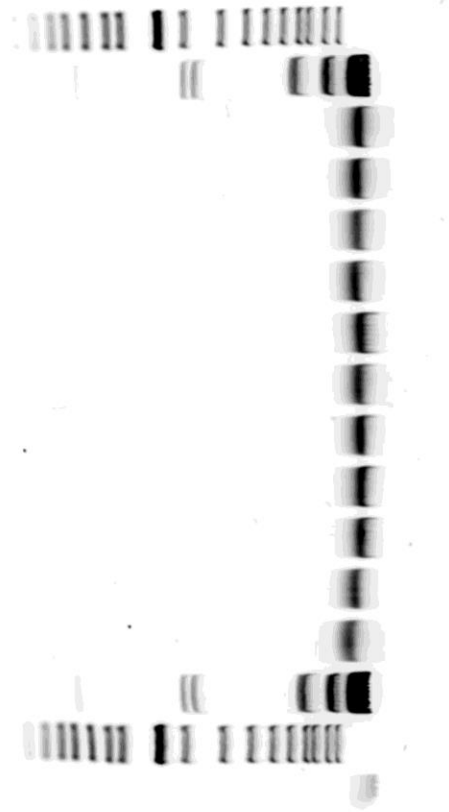


Note: the Fragment Analyzer traces of sample 7 were not analyzed because the digestion QC baseline can't be adjusted



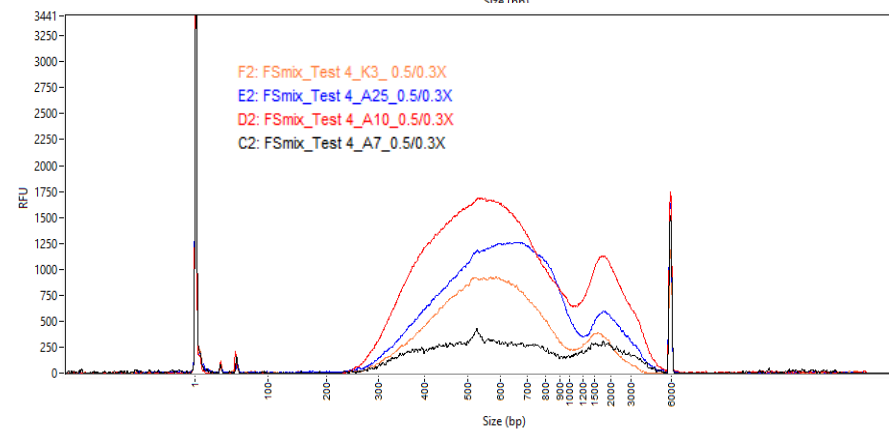
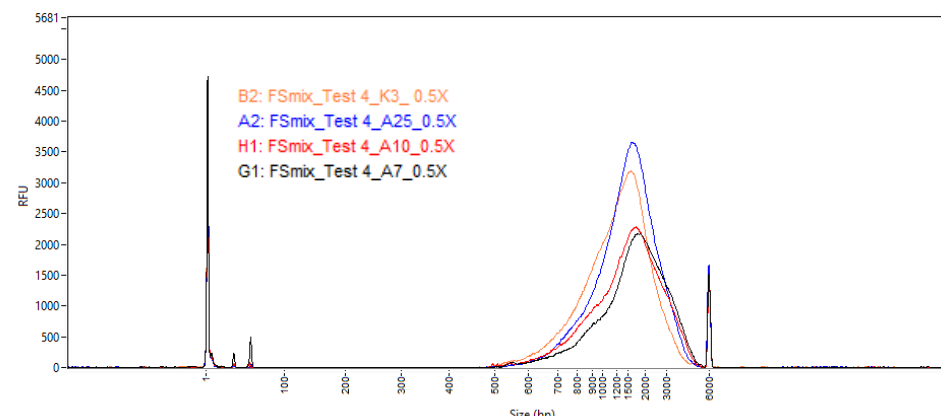
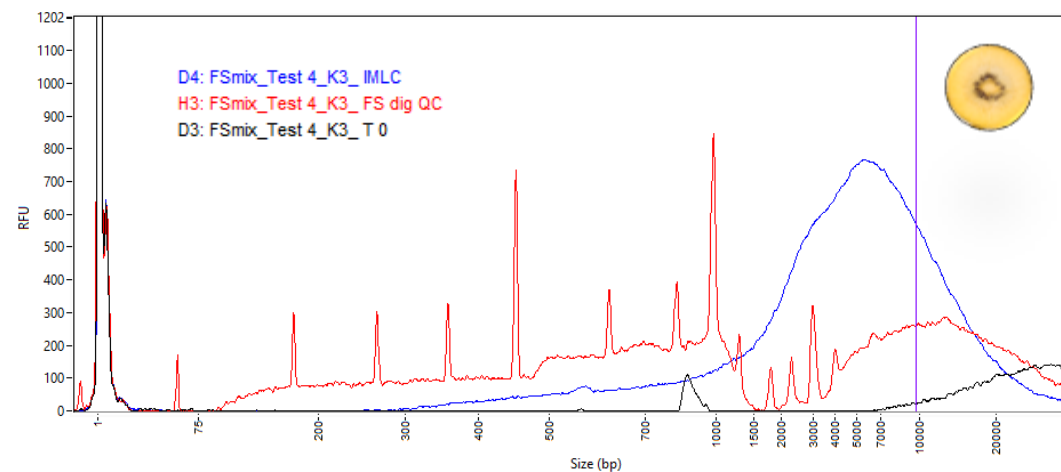
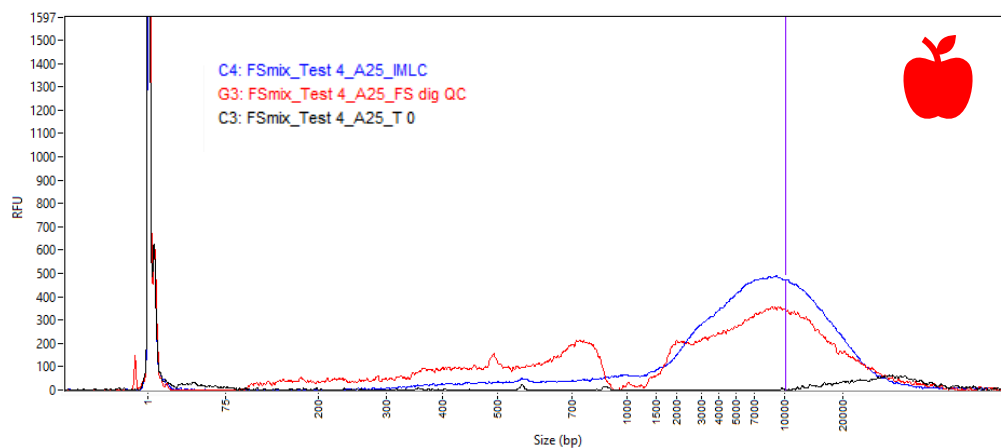
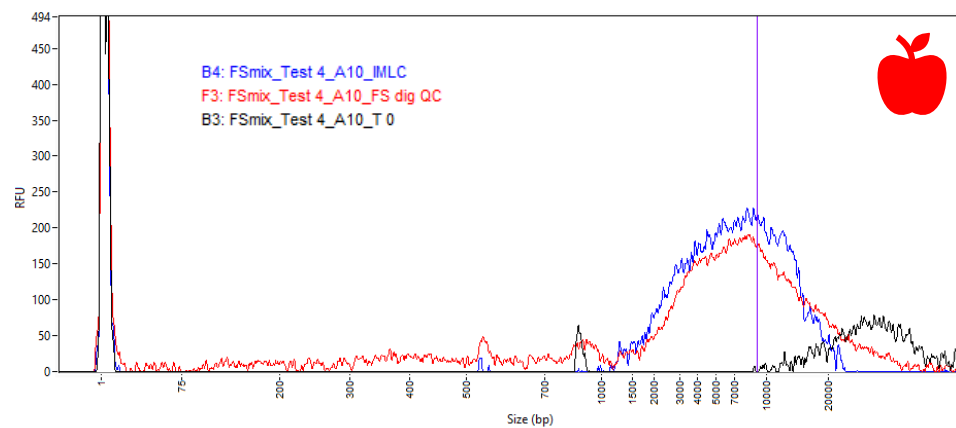
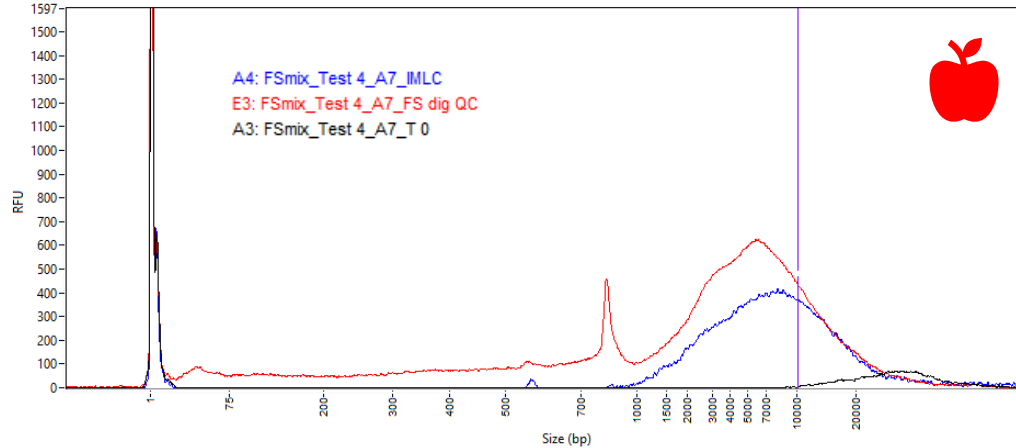
FSmix **Test 4** Lysate QC

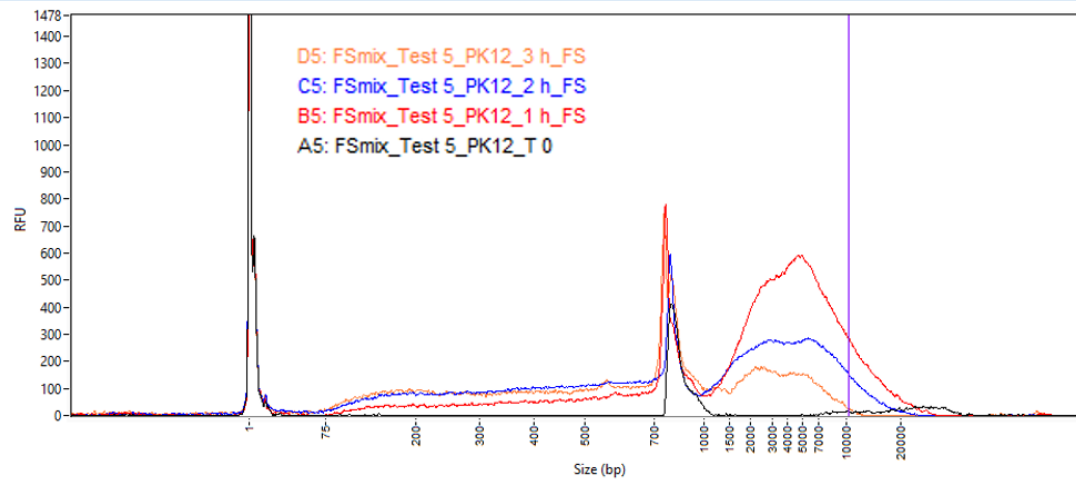
3/08/2026



- 1 kb+
- Lambda Hind III
- **A7**
- **A10**
- A11
- A19
- A20
- A21
- A23
- **A25**
- A26
- A27
- **Kiwifruit 3**
- Lambda Hind III
- 1 kb+
- Lambda 48.5 kbp

200 ng/lane, 0.75% agarose 1X TAE, 120 V, 40 min, SybrSafe

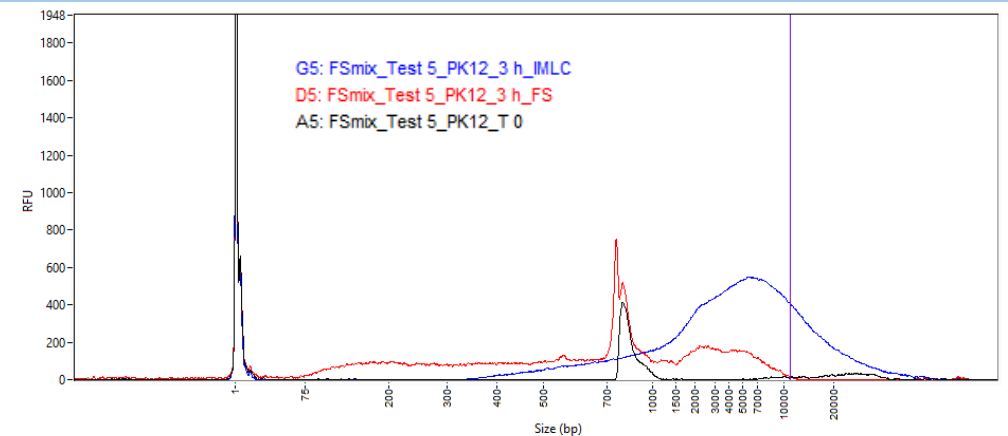
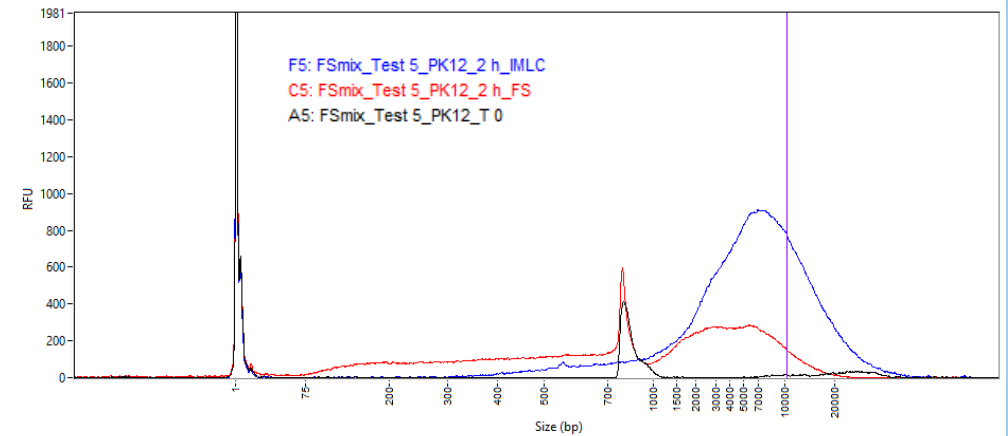
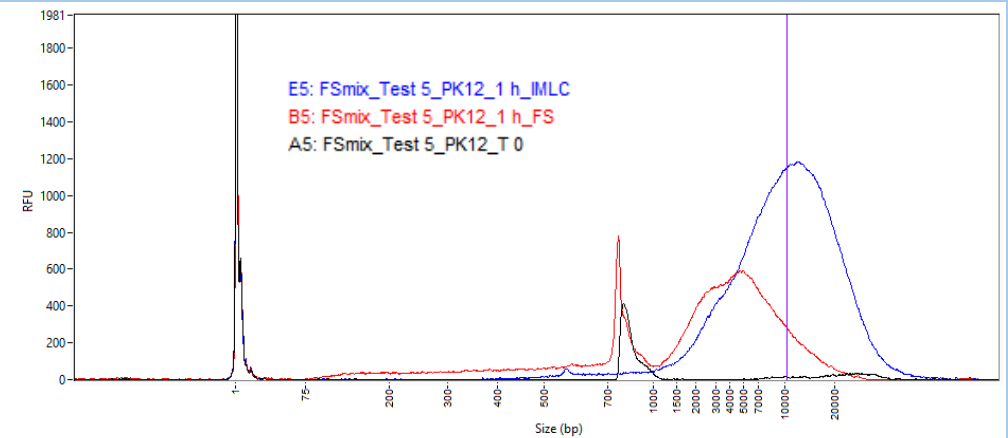




FSmix digestion time course

- A kiwifruit crosslinked chromatin lysate was used for **Test 5**
- About 400 ng of chromatin bound to magnetic beads was used for starting the FSmix digestion (2 μ L enzyme mix, 4 μ L FSmix buffer, total volume 20 μ L, 37°C, 1250 rpm)
- The reaction was stopped by placing the tube on the magnet, and washing the chromatin with Wash buffer (Tris pH 8, Tween 20) + 1 mM EDTA, twice, 500 μ L each wash
- Ten percent of the reaction was used for QC, the rest went to bridge ligation and intramolecular ligation

Before and after FSmix digestion, followed by intramolecular ligation for each time point

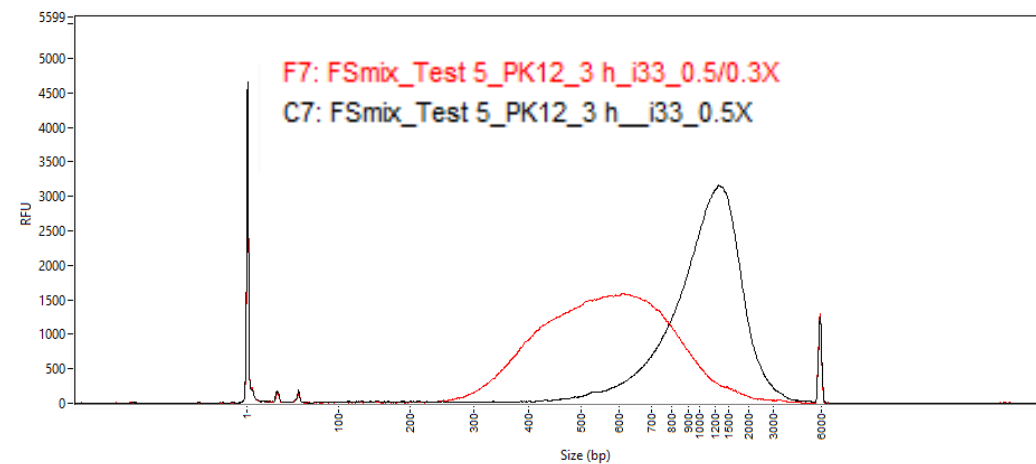
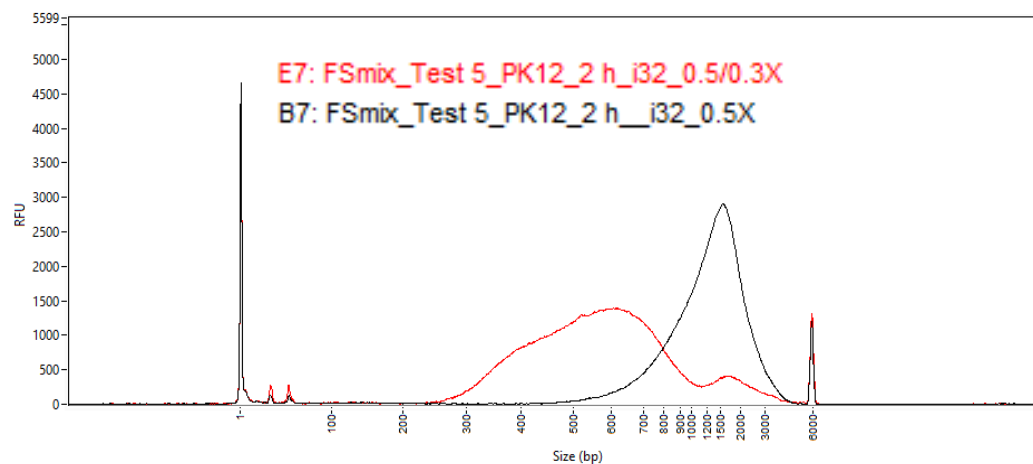
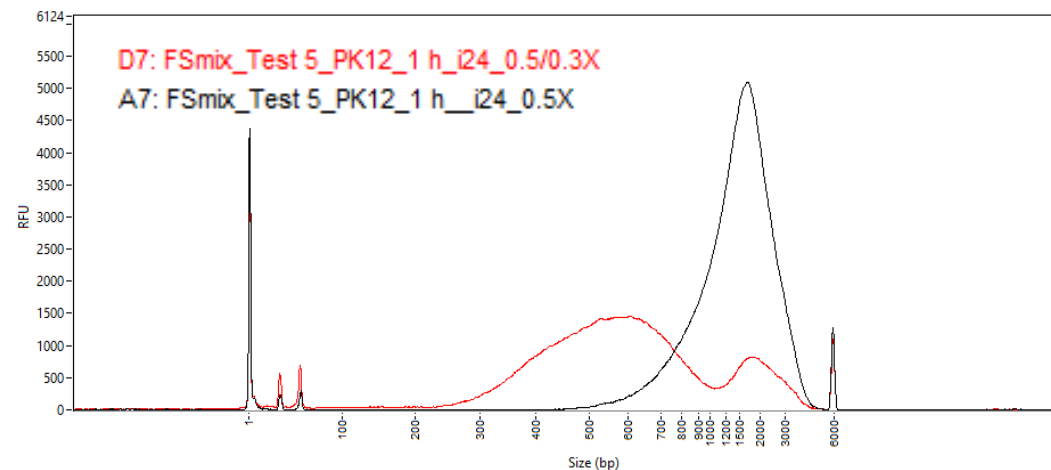


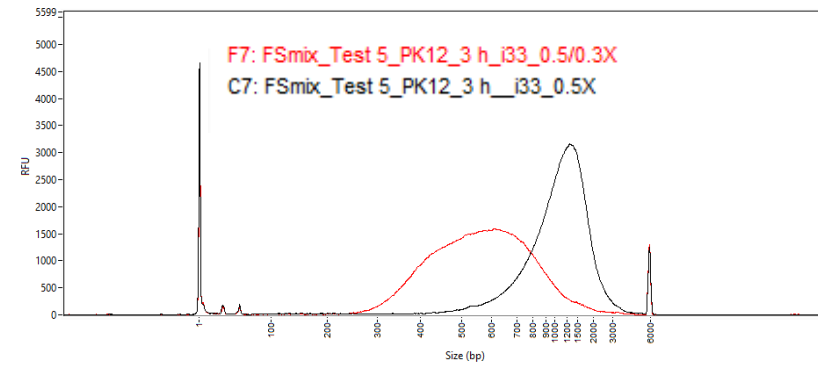
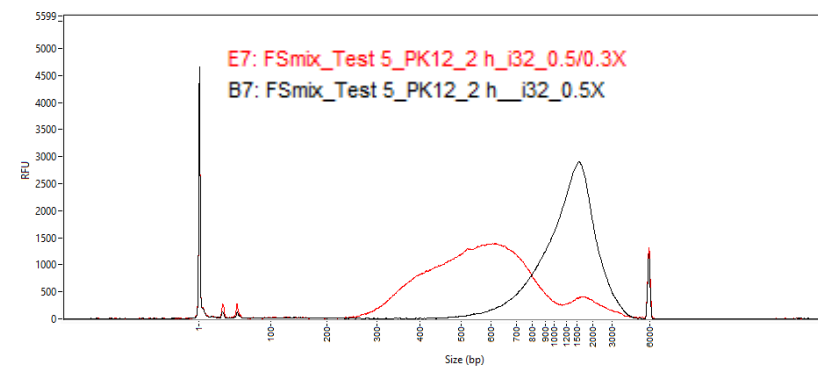
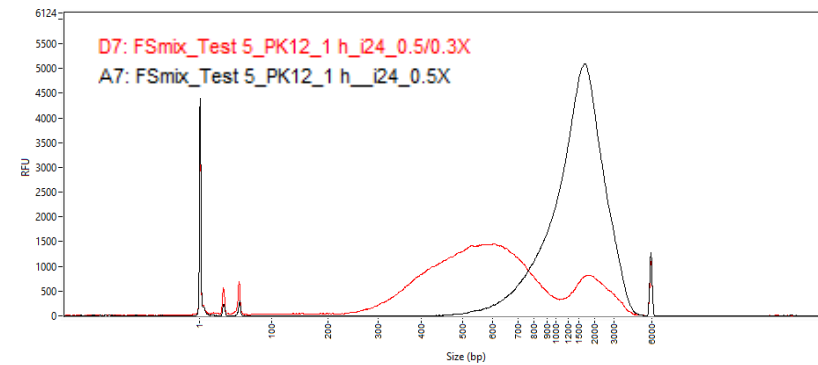
Comparison of the high and lower cut of the amplified library

The PCR reaction volume (25 μ L) was adjusted to 100 μ L before adding 50 μ L AMPure XP beads. After binding and magnet separation, the supernatant was transferred to a new tube containing 30 μ L AMPureXP beads. Both bead fractions are cleaned up and eluted in TE pH 7.5 buffer

The ratio of amount of amplified library between the two fractions might be an indication of how well the FSmix digested the chromatin

	Digestion time	Total, ng	0.5X : 0.5/0.3X
0.5X	1 h	268	
	2 h	322	
	3 h	292	
0.5/0.3X	1 h	60	4.4 : 1
	2 h	127	2.5 : 1
	3 h	298	0.97 : 1





Sample Name	% Duplication	Reads After Filtering	GC content	% PF	% Adapter
Red19_FS1H	1.70%	8.7M	36.90%	100.00%	2.30%
Red19_FS2H	1.30%	15.1M	36.80%	100.00%	1.90%
Red19_FS3H	0.50%	14.6M	36.90%	100.00%	1.60%

FS digest	IMLC, ng	IMLC-Illumina, ng	Yield Illumina	SAMB unbound, ng	% IMLC-Illumina SAMB bound
1 h	362	264	73%	166.4	37%
2 h	344	290	84%	159.2	45%
3 h	210	240	114%	116	52%